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POWERING DELHI
EMPOWERING CONSUMERS

Great Place To Work. Certified DEC 2024-DEC 2025 INDIA

Kilokari Battery Energy Storage System

Case Study Subject

BSES: Silver Jubilee year of serving the National Capital

BSES Rajdhani Power Limited | BSES Yamuna Power Limited

A Grid Serving 2+ Crore Lives

52 Lakh

Consumers

BRPL 31.7L · BYPL 20.3L — 2 of every 3 Delhiites

5,833 MW

2026 Peak Demand

BRPL 3,906 · BYPL 1,927 — more than all other metros combined

3,518 MW

Long-Term RE

BRPL 2,531 · BYPL 987 (excl. hydro) — towards a Green DISCOM

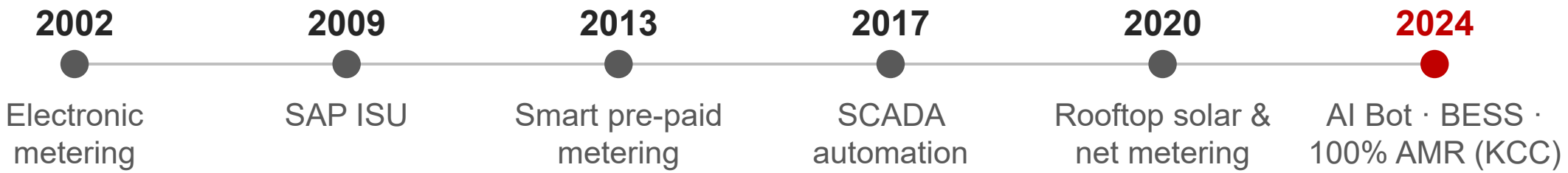
18,000+

Jobs Created

Direct and indirect jobs across the network

AT&C Loss Trajectory

AT&C losses are well below the national average of 15.4% (PFC, FY22-23), driven by metering, SCADA, GIS/OMS analytics, smart pre-paid metering and AI-enabled operations — the digital backbone for BESS and DER integration.



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Industry Challenge

- India targets 500 GW of non-fossil capacity by 2030; ~308 GW already installed (Sept 2026)
- Peak power demand already reached ~270 GW till now in 2026, with cities like Delhi seeing fast load growth
- Dense urban grids face rising peaks and limited space for network expansion
- Utility-owned, distribution-connected BESS in ancillary markets remains largely untested in India
- Peak loads are concentrated in short periods: the top 10% and 20% load levels occur for only 1.45% and 6.53% of the year, respectively

500 GW

India's non-fossil capacity target by 2030 (~308 GW installed, Sept 2026)

~270 GW

India's peak power demand reached in 2024

Grid Flexibility Gap

Rising variable renewable energy creates volatility that traditional infrastructure cannot absorb

Urban Space Constraints

Dense city environments demand compact, high-impact energy solutions

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Kilokari BESS: Built and Commissioned in 2 Years

South Asia's first commercially deployed 20 MW / 40 MWh regulated BESS at 11 kV



Capacity	Battery
20 MW / 40 MWh at 11 kV	Lishen LFP, 2 cycles/day
PCS	**Transformer**
Fimer 2.5 MVA × 12	3 × 7.4 MVA, Raychem
Concession	**Developer**
12 years (8,760 cycles)	Kilokari BESS Pvt. Ltd.

Delivery milestones

May 2023	Jul 2023	Dec 2023	May 2024	Dec 2024	Apr 2025
Feasibility assessment	DERC in-principle approval	BESS Agreement signed	Tariff adopted by DERC	FAT of battery containers	Project commissioned

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5-Layer Fire Safety System

Layered protection built to international battery-storage safety standards

1. **Cell-Level Suppression**
   Early Thermal Runaway Detection.

2. **Aerosol Extinguishing**
   Fixed aerosol fire extinguishing system per NFPA 2010.

3. **Active Ventilation**
   Deflagration venting per NFPA 68 / 69 explosion-proof standards.

4. **Enclosure Fire Rating**
   2-hour fire resistance (ASTM E119 / UL 263) vs typical 1-hour norm.

5. **Fire Water Spray and Puff Panel**
   Avoid Reignition and additional 2 Hrs. Fire protection

Standards Compliance

- ✓ NFPA 855 — Installation
- ✓ NFPA 68 & 69 — Explosion Venting
- ✓ NFPA 720 — CO Detection
- ✓ NFPA 2010 — Aerosol Suppression
- ✓ UL 9540 / 9540A
- ✓ UL 864, 268, 521, 1203
- ✓ IEC 62933-5-2 — Functional Safety

Reduced Clearance Design

Safe spacing cut to 3 ft (0.9 m) vs the standard 10 ft (3 m).

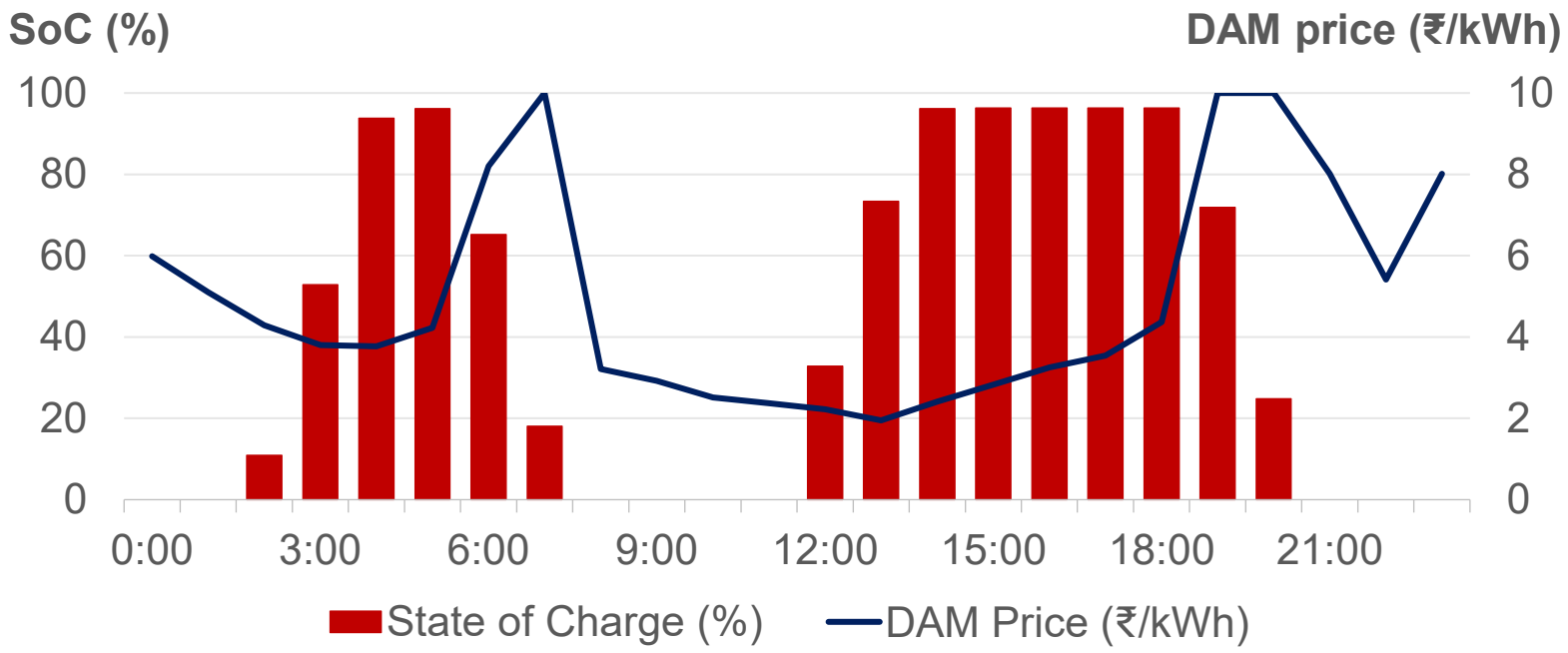
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BESS Performance till Sept 2026

Operating since April 2025 — cumulative performance and a typical dispatch day

36.08 MU	32.20 MU	89.24%	805
Charging energy	Discharging energy	Round-trip efficiency	Cycles completed

Real dispatch — 4 April 2025 (charge at low DAM prices)



Four revenue streams

1. Energy arbitrage
2. Reactive power support
3. Blackout support
4. Ancillary services (SRAS)

One asset: all four run on the same 20 MW / 40 MWh capacity.

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Grid Impact and a National First

Peak shaving at the local grid and India's first ancillary-services test for a BESS

~38%

Peak load reduction
Kilokari BRPL Grid

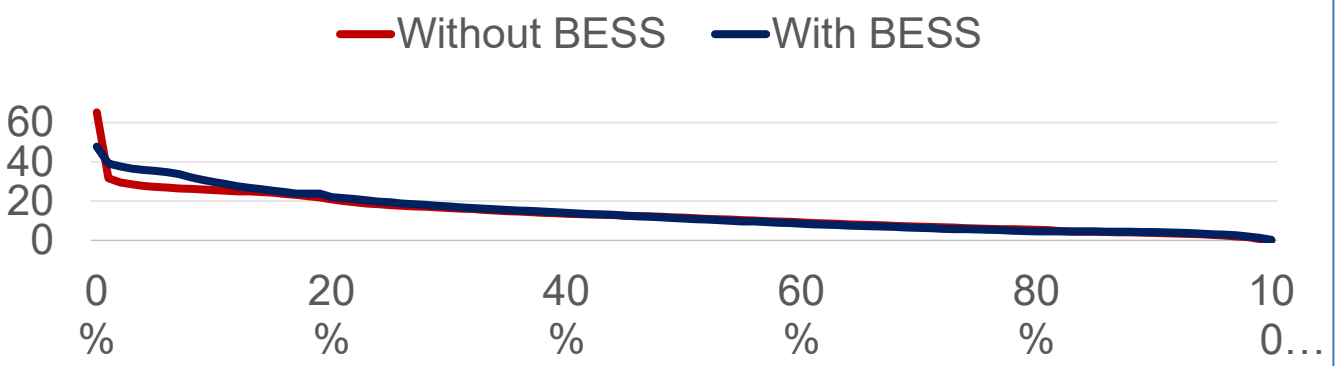
~8%

Peak load reduction
IP DTL Grid Station

8,748 MW

Delhi peak, 29 Jun 2026
Charged on solar hours, discharged at night peak

Load duration curve: pre vs post BESS (20 MW × 2 hrs)



Metric	Without	With	Change
Peak load (MW)	65.23	**47.75**	**−27%**
Load factor	24.45%	**34.04%**	**+9.6 pts**
Utilisation	114.44%	**83.78%**	**−30.7 pts**

National first
13 May 2025

India's first BESS tested for Secondary Reserve Ancillary Services: SRAS-up (discharge) and SRAS-down (charge) on AGC signals.

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Kilokari: India’s First BESS Tested for Ancillary Services

Test completed: 13 May 2025

Kilokari became the first energy storage system in India tested for Secondary Reserve Ancillary Services (SRAS) — proving regulated, DISCOM-owned storage can deliver grid balancing alongside conventional generation.

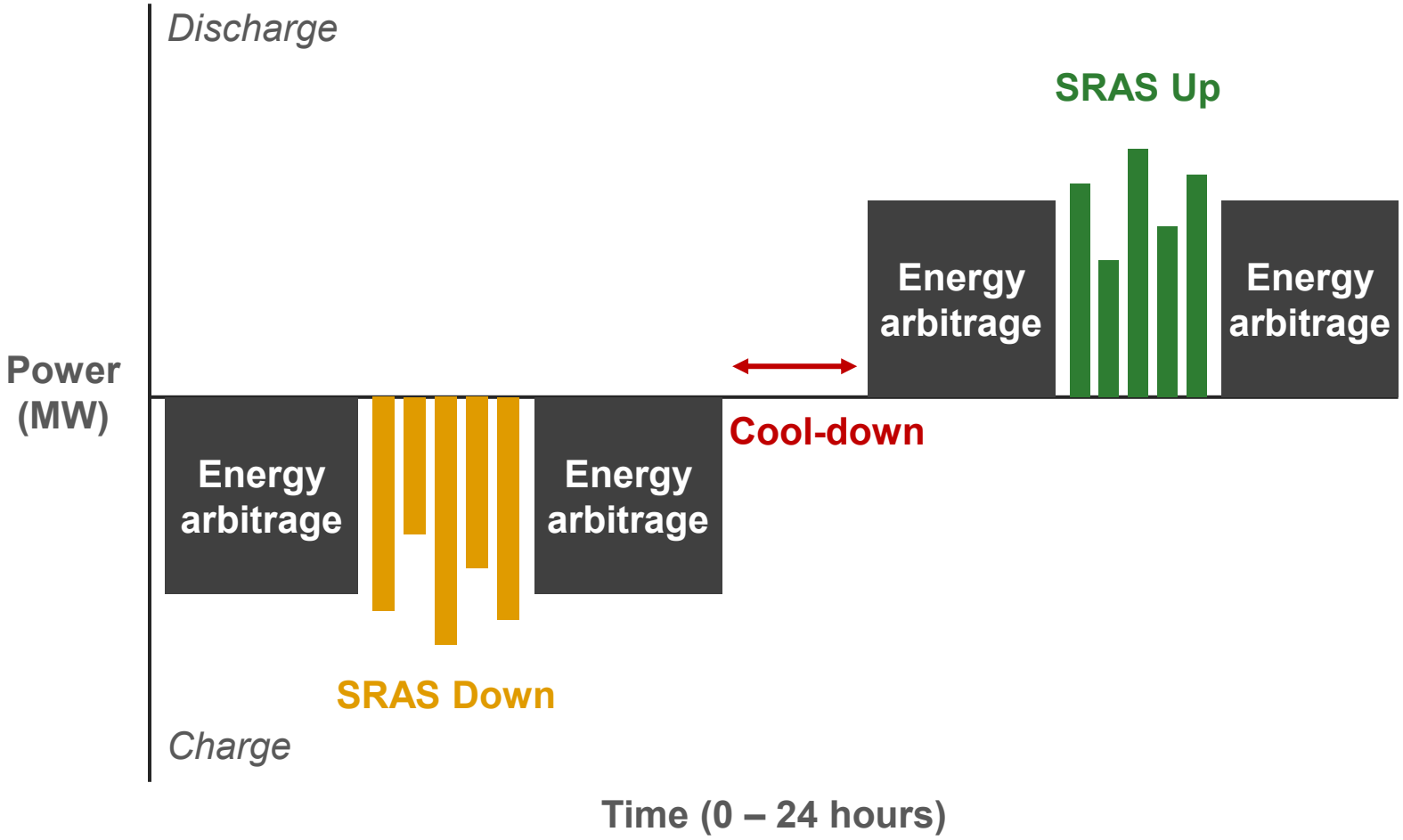
SRAS Up	Discharge on AGC signals during under-frequency events
SRAS Down	Charge to absorb excess generation during over-frequency
Response basis	Automatic Generation Control (AGC) signal-driven dispatch



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Illustration of Value Stacking for Kilokari BESS

Use-case: Energy arbitrage + SRAS



Energy arbitrage

Charge at low, discharge at high DAM prices

SRAS Down

Charging on AGC signals

Cool-down period

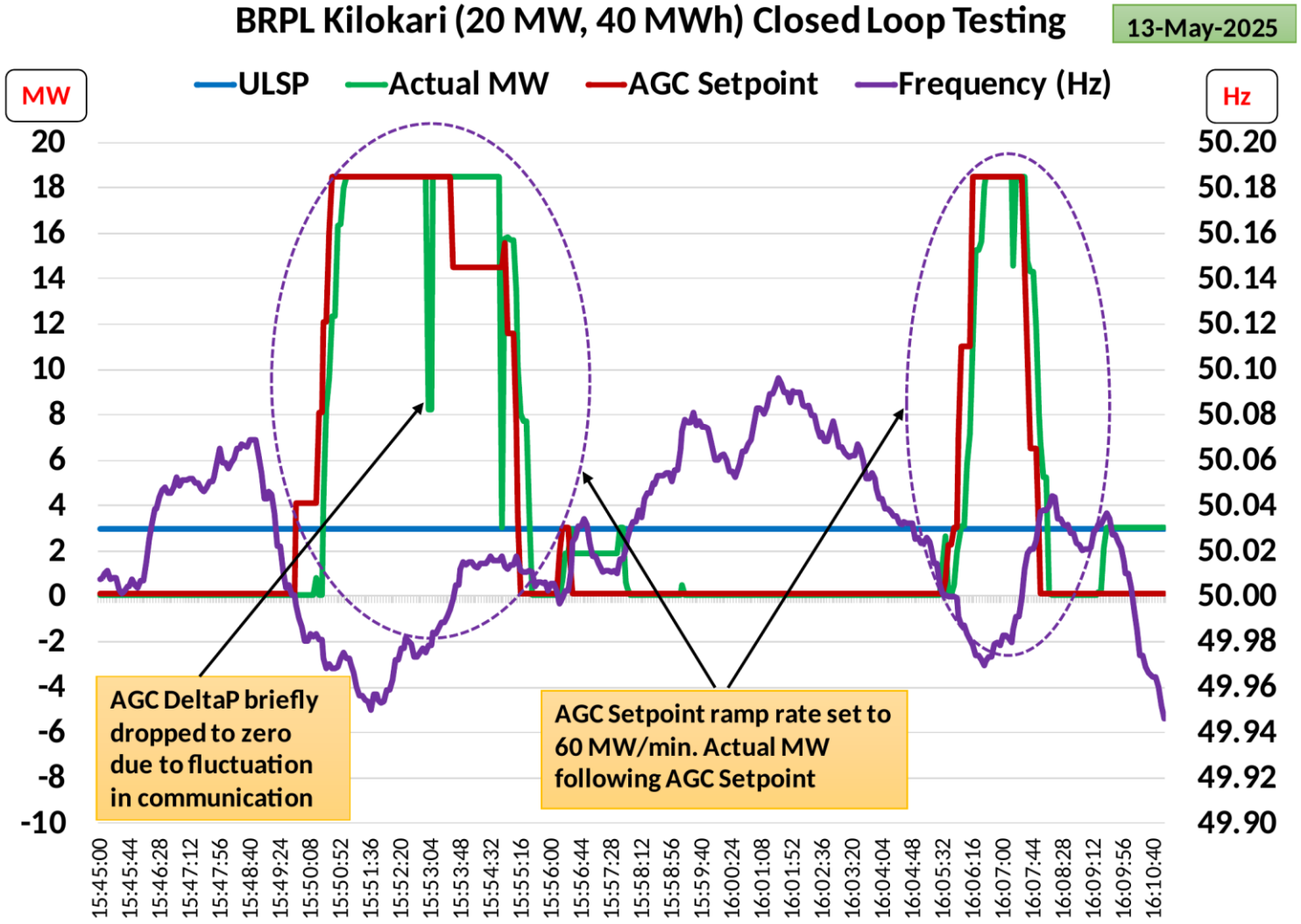
Rest between cycles

SRAS Up

Discharging on AGC signals

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SRAS participation : Illustration of BESS discharge based on AGC signals



What the test shows

13 May 2025

Closed-loop test, 20 MW / 40 MWh

Up to ~18.5 MW

Discharged when frequency fell below 50 Hz

60 MW/min

AGC setpoint ramp; actual MW followed the setpoint

Brief dip

AGC ΔP dropped to zero due to a communication fluctuation

Unlocking Full BESS Participation in SRAS Markets

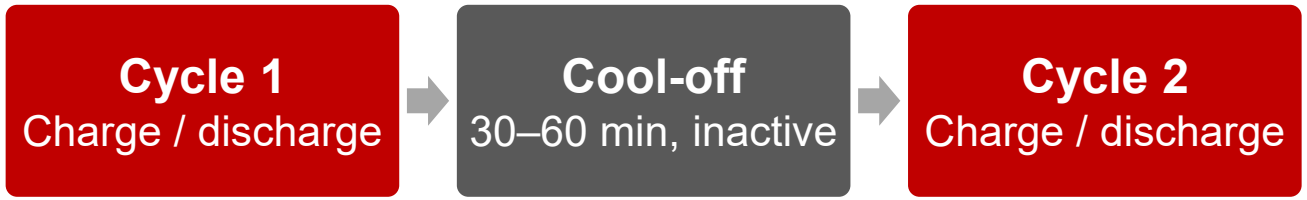
Operating experience at Kilokari points to five regulatory changes needed for fair participation of storage in ancillary service markets:

1. Recognise Electrical Energy Storage (EES) as a separate asset class alongside generators
2. Revise clauses to compensate SRAS-down (charging) delivered by storage systems
3. Introduce cost-reflective settlement accounting for round-trip efficiency and SoC limits
4. Permit dual bidding for SRAS-up and SRAS-down to reflect operational asymmetries
5. Clarify that charging energy for ancillary services is not a market deviation

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Concept of cool-off period in India

A **“cool-off period”** is a specified duration (commonly 30 minutes to an hour) when the battery must remain inactive — neither charging nor discharging — between two operational cycles.



- The concept is globally understood; the term “cool-off period” is more common in Indian battery operation contexts and some Asian markets.
- US and European RFPs and operating guidelines phrase it differently — “rest”, “idle” or “thermal stabilisation” periods.

Related terminology used internationally

India

- Cool-off period

United States

- Rest Interval
- Idle Time
- Standby Period

Europe

- Battery Rest Period
- Thermal Stabilization Interval
- Non-Operational Duration

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Thank You

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